VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)

Department of Computer Engineering



Project Report on

Conversational Model for Indian Sign Language (ISL)

In partial fulfillment of the Third Year (Semester–VI), Bachelor of Technology (B.Tech.) Degree in Computer Engineering at the University of Mumbai

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(2025-26)

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

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Certificate of Approval

This is to certify that ***Lakhan Gurubaxani, Saniya Jeswani, Vidisha Kataria & Ishjyot Kaur*** of Third Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on "***Conversational Model for Indian Sign Language (ISL)***" as a part of the coursework of CAPSTONE PROJECT - I for Semester-VI under the guidance of ***Mrs. Lifna C.S.*** in the year 2025-26.

_____

Date

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Project Mentor	Head of the Department	Principal
Mrs. Lifna C S	Dr. Mrs. Nupur Giri	Dr. J. M. Nair

ACKNOWLEDGEMENT

We are thankful to our college Vivekanand Education Society's Institute of Technology for considering our project and extending help at all stages needed during our work of collecting information regarding the project.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR CAPSTONE PROJECT

Learners will be to:-

Course Outcomes	Description of the Course Outcomes
CO1	Identify and define a real-world problem through systematic requirement analysis and investigation.
CO2	Conduct a thorough literature review and benchmarking to propose innovative solutions.
CO3	Develop a structured project plan with well-defined milestones, timelines, and risk mitigation strategies.
CO4	Design and implement a working prototype using appropriate engineering tools and methods.
CO5	Analyze experimental results, validate the solution, and optimize performance based on data.
CO6	Effectively document and present the project outcomes, demonstrating professional ethics, teamwork, and project management skills.

ABSTRACT

Communication barriers remain a significant challenge for hearing-impaired individuals, particularly in regions where **Indian Sign Language (ISL)** is not widely understood. The lack of interpreters in educational institutions, workplaces, healthcare centers, and public services often results in social isolation and reduced independence.

This project proposes an **Indian Sign Language Gesture Recognition and Smart Conversation System** designed to translate **hand gestures into text and speech** in real time. Using computer vision and deep learning techniques, the system will accurately detect gestures through a camera and convert them into understandable outputs.

Additionally, the system is designed to **improve accessibility** and **usability** through **efficient** and **user-friendly interactions**. By leveraging modern technologies, the proposed solution promotes inclusive communication, empowers hearing-impaired individuals, and supports their active participation in society.

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Chapter 1: Introduction

This chapter provides an overview of the project, its motivation, limitations of existing systems, problem definition, relevance, and methodology used in the proposed system.

1.1 Introduction to the Project

Communication is a fundamental necessity for human interaction. However, hearing-impaired individuals face significant challenges due to the limited understanding of Indian Sign Language (ISL) among the general population.

The proposed project aims to develop a **Conversational Model for ISL** that recognizes hand gestures using computer vision and deep learning techniques and converts them into text and speech in real time.

1.2 Motivation for the Project

- Lack of ISL interpreters in public spaces
- Communication barriers in education and workplaces
- Need for inclusive and accessible technology
- Advancement in AI enabling real-time gesture recognition

1.3 Drawback of the Existing System

- High cost of interpreter services
- Lack of real-time automation
- Limited portability
- Inconsistent accuracy in current systems
- Limited Translator

1.4 Problem Definition

Hearing-impaired individuals face communication challenges due to the lack of ISL understanding among people. Existing systems fail to provide real-time, affordable, and scalable solutions.

1.5 Relevance of the Project

- Promotes inclusive communication
- Enhances independence of users
- Applicable in education, healthcare, and workplaces

1.6 Methodology Used

- **Dataset Collected:** ISign Dataset.
- **Data Preprocessing:** Perform data cleaning, labeling, resizing, and normalization to improve model performance.
- **Model Development:** Implement deep learning algorithms such as Convolutional Neural Networks (CNN) or MediaPipe-based hand tracking for accurate gesture detection.
- **Training and Validation:** Train the model using prepared datasets and evaluate it using performance metrics like accuracy and precision.
- **Gesture Translation Module:** Convert recognized gestures into readable text and generate speech output using text-to-speech technology.
- **System Testing:** Test the system under varying lighting conditions, backgrounds, and user scenarios to ensure reliability.
- **Deployment:** Develop a mobile or web-based interface to provide a user-friendly experience.

Chapter 2: Literature Survey

This chapter reviews existing research, references, expert interactions, and patents relevant to the proposed system.

2.1 Research Papers

Paper 1 : Sign language recognition systems: A decade systematic literature review (2021)

Abstract: This paper presents a comprehensive review of sign language recognition (SLR) systems developed over the past decade. It analyzes various techniques used for recognizing sign language, including image processing, machine learning, and deep learning approaches such as CNNs and RNNs. The study compares sensor-based and vision-based systems, highlighting their advantages, limitations, and performance metrics. It also discusses challenges such as dataset availability, signer variability, real-time processing, and environmental constraints. The review identifies trends in the field and suggests future directions for improving accuracy, scalability, and usability of SLR systems.

Inference: The paper concludes that deep learning-based approaches significantly outperform traditional methods in sign language recognition. However, there is still a lack of standardized datasets and real-time robust systems. Future research should focus on creating large, diverse datasets and improving real-time recognition systems to enhance accessibility for the deaf community.

Paper 2 : ISLTranslate: Dataset for translating Indian sign language (2023)

Abstract: This paper introduces ISLTranslate, a dataset specifically designed for translating Indian Sign Language (ISL) into text. The dataset includes annotated video samples of various ISL gestures, capturing different signers, lighting conditions, and backgrounds to improve generalization. The authors also propose baseline models using deep learning techniques for translation tasks. The study emphasizes the importance of domain-specific datasets for improving translation accuracy and bridging communication gaps between deaf and hearing communities.

Inference: The dataset plays a crucial role in advancing ISL translation systems. It highlights that the availability of high-quality, diverse datasets directly impacts model performance. The research indicates that future systems can achieve better translation accuracy by leveraging larger datasets and advanced deep learning architectures.

Paper 3 : Sign language generation system based on Indian sign language grammar (2020)

Abstract: This paper focuses on generating sign language from text based on Indian Sign Language grammar rules. It proposes a system that converts spoken or written language into ISL by incorporating linguistic structures such as word order, facial expressions, and gestures. The system uses rule-based and computational techniques to ensure grammatical correctness in generated signs. It aims to create more natural and understandable sign language outputs for users.

Inference: The study shows that incorporating linguistic and grammatical rules significantly improves the quality of generated sign language. However, rule-based systems may lack flexibility and scalability. Future work should integrate AI-based approaches with grammar rules to produce more natural and adaptive sign language generation systems.

Paper 4 : A Sociolinguistic study of the use of Indian Sign Language (2016)

Abstract: This paper examines the sociolinguistic aspects of Indian Sign Language (ISL), including its usage, variations, and cultural significance. It explores how ISL differs across regions, communities, and educational backgrounds. The study highlights the lack of standardization and the impact of social factors on language use. It also discusses the importance of promoting ISL for better inclusion and communication within the deaf community.

Inference:The paper concludes that ISL is highly diverse and influenced by social and regional factors. The lack of standardization poses challenges for developing uniform recognition and translation systems. Future efforts should focus on standardizing ISL and increasing awareness to support technological advancements and inclusivity.

2.2 Books / Articles / Newspaper

This project is based on a review of multiple sources including:

- Articles and research papers related to Artificial Intelligence and Computer Vision, focusing on object detection, image recognition, and deep learning techniques.
- Studies on NLP-based communication systems, covering text processing, language models, and human-computer interaction.
- Online technical documentation, blogs, and educational resources used to understand implementation details and practical applications.

Detailed references for all sources are provided in the **References** section at the end of the report.

2.3 Interaction with Domain Experts

- Faculty guidance
- Insights on usability and deployment

2.4 Patent Search

Patent 1 : Higaki, Nobuo, Yuichi Yoshida, and Kikuo Fujimura. "Gesture recognition system." U.S. Patent No. 6,804,396. 12 Oct. 2004.

Title: Gesture Recognition System (2004)

Summary: This patent describes a camera-based system that captures human hand or body gestures in real time and processes them using image processing techniques. The system identifies specific gesture patterns by analyzing motion, shape, and position, enabling touchless interaction with devices. It can be applied in areas such as human-computer interaction, gaming, and assistive technologies.

Patent 2 : Starner, Thad, Joshua Weaver, and Alex Pentland. "Real-time American sign language recognition using desk and wearable computer based video." U.S. Patent No. 6,975,090. 13 Dec. 2005.

Title : Sign Language Translation Device (2005)

Summary: This invention focuses on translating sign language gestures into spoken or text-based output. It uses sensors or cameras to capture hand movements and applies recognition algorithms to interpret the gestures. The system then converts the recognized signs into audible speech or readable text, helping bridge communication gaps between hearing-impaired individuals and others in real-time.

Patent 3 : Patent 3 : Yoo, Myung-Jin, et al. "Automated gesture identification using neural networks." U.S. Patent No. 11,087,488. 10 Aug. 2021.

Title : Automated Gesture Identification Using Neural Networks (2021)

Summary: This patent introduces an AI-based communication system that leverages machine learning models to recognize and interpret gestures more accurately over time. By training on large datasets of gesture inputs, the system improves its prediction and translation capabilities. It can adapt to different users and gesture variations, making it suitable for advanced assistive communication, smart interfaces, and inclusive technology solutions.

Chapter 3: Requirement of Proposed System

This chapter defines system requirements and constraints.

3.1 Functional Requirements

- Capture gestures via camera
- Recognize ISL gestures
- Convert gestures to text
- Generate speech output

3.2 Non-Functional Requirements

- High accuracy
- Low latency
- User-friendly interface
- Scalability

3.3 Constraints

- Lighting conditions
- Dataset limitations
- Hardware limitations

3.4 Hardware & Software Requirements

Hardware:

- Laptop (8GB RAM)
- Camera
- GPU (optional)

Software:

- Python
- OpenCV / MediaPipe
- TensorFlow / PyTorch
- Android Studio / Web Framework

3.5 Techniques Utilized

- Deep Learning (CNN+LSTM)

- Computer Vision
- NLP

3.6 Tools Utilized

- Python
- TensorFlow
- OpenCV
- MediaPipe

3.7 SDGs Mapped

- **SDG 9:** Industry, Innovation, and Infrastructure
- **SDG 10:** Reduced Inequalities

Justification:

The proposed system promotes innovation by applying AI and computer vision technologies (SDG 9) while improving accessibility for hearing-impaired individuals, thereby reducing communication barriers and social inequalities (SDG 10).

Chapter 4: Proposed Design

This chapter explains system architecture and design.

4.1 Block Diagram

The block diagram represents the overall workflow of the proposed ISL conversational system. It begins with dataset collection and preprocessing, followed by feature extraction and deep learning-based model development for gesture recognition. The recognized gestures are then converted into text and speech, and the system is finally deployed as a user-friendly application for real-time communication.

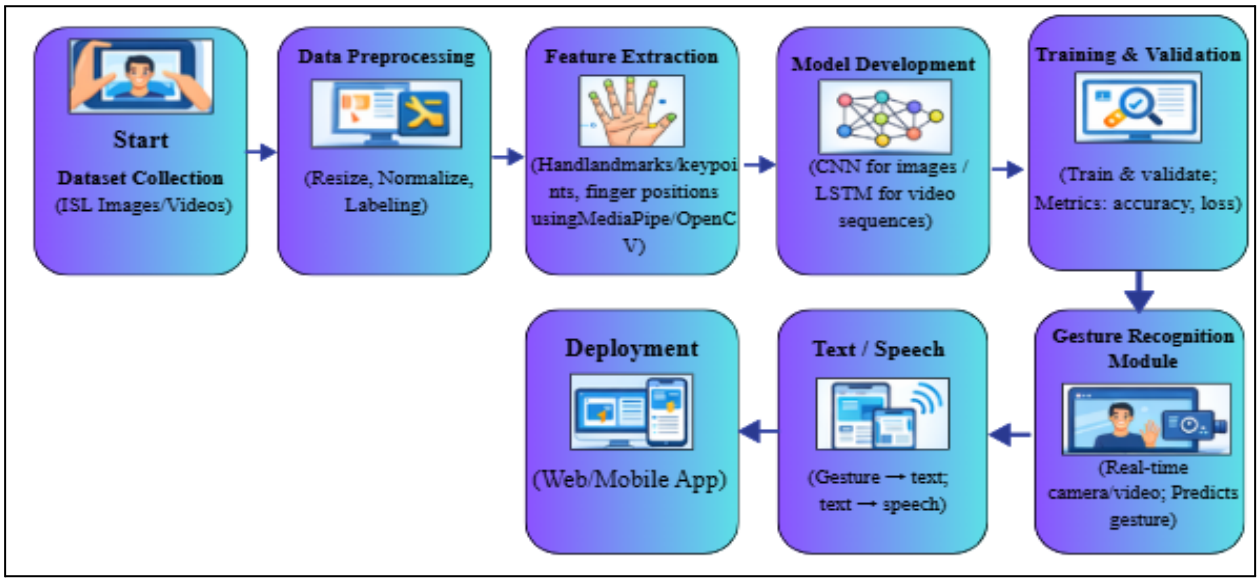


Figure - 1

4.2 Modular Diagram

The modular diagram illustrates the system divided into functional components such as input processing, feature extraction, temporal modeling, and translation. Each module performs a specific task, from extracting hand landmarks and learning gesture sequences to converting them into meaningful text. This modular approach improves system scalability, maintainability, and overall performance.

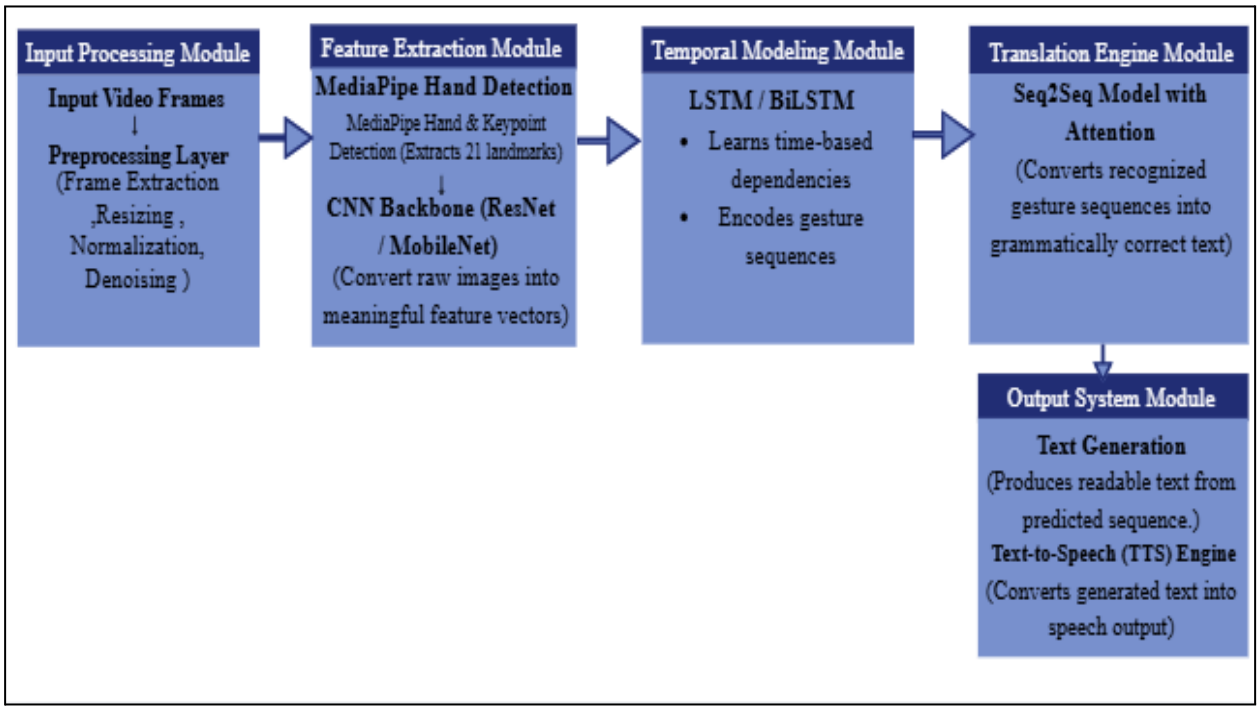


Figure - 2

4.3 System Design

Includes:

- Use Case Diagram

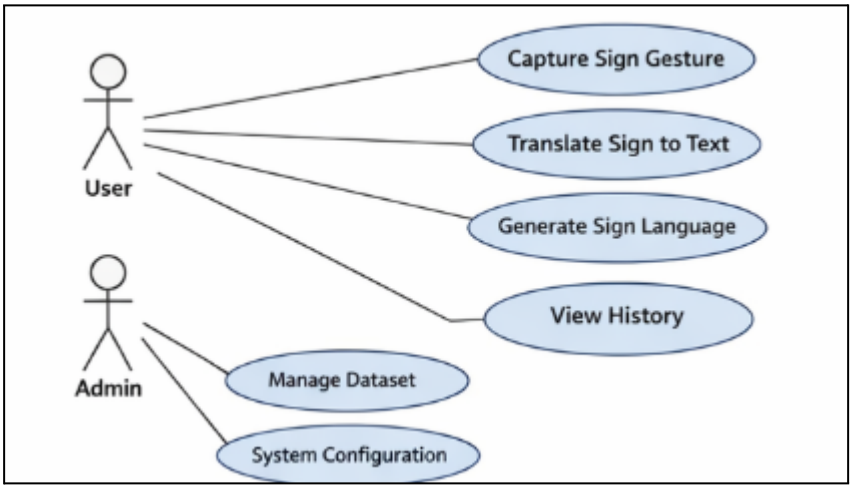


Figure - 3

This diagram illustrates the interactions between two actors — the User and the Admin — and the system. The User can capture sign gestures, translate signs to text, generate

sign language, and view history, while the Admin manages the dataset and handles system configuration to ensure smooth operation.

- Flowchart

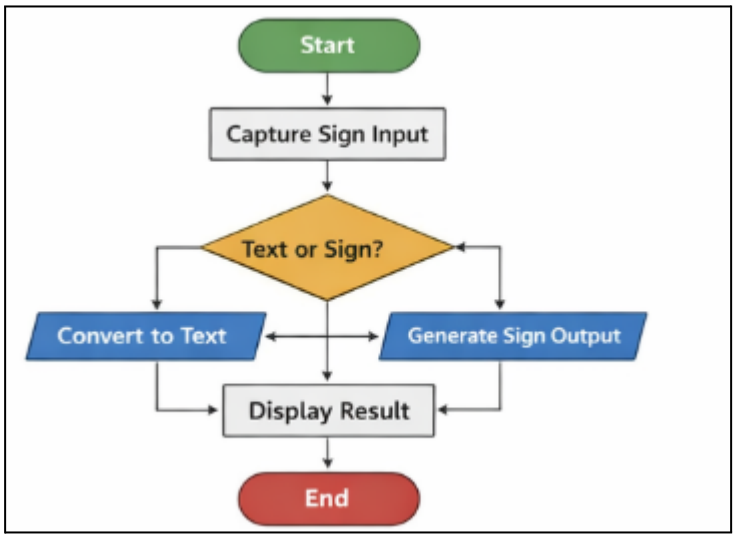


Figure - 4

Depicts the core processing logic of the ISL system. It begins with capturing sign input, followed by a decision point that determines whether the input should be converted to text or generate a sign output, and finally displays the result to the user.

- ER Diagram

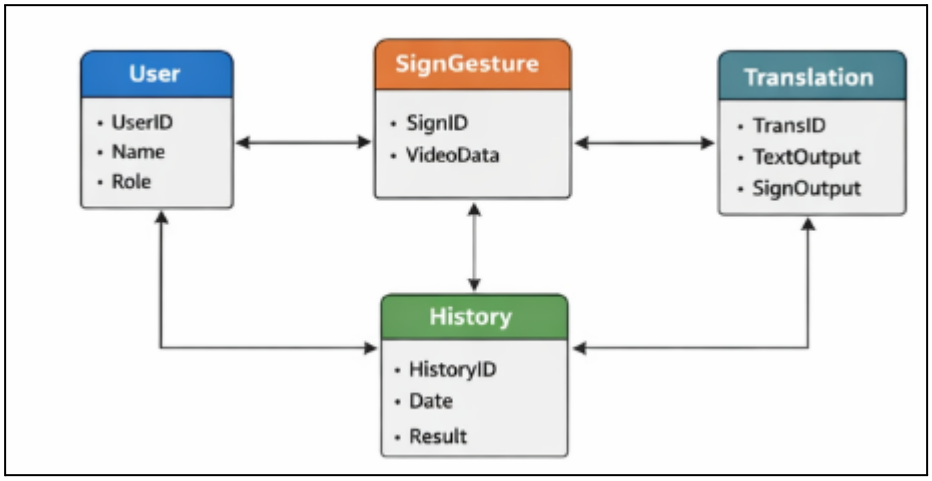


Figure - 5

The ER diagram represents the database structure of the system, showing four entities: User, SignGesture, Translation, and History. It captures relationships such as a user performing sign gestures that get translated into text or sign output, with results being stored in the history for future reference.

- Activity Diagram

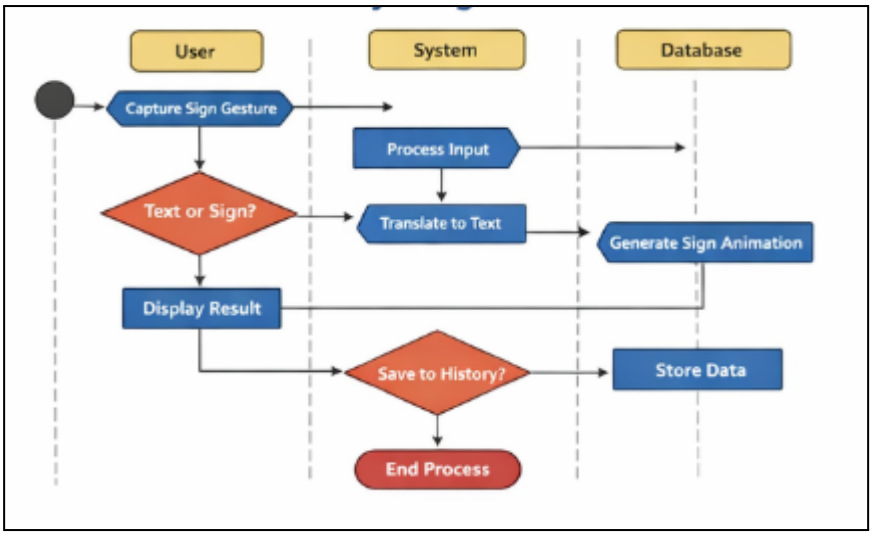


Figure - 6

Models the dynamic workflow across three swimlanes — User, System, and Database. It shows how a captured sign gesture is processed and translated by the system, optionally generating a sign animation, displaying the result to the user, and saving the session data to the database if required.

4.4 Algorithms Utilized

- CNN for image classification
- MediaPipe hand tracking
- LSTM for sequential data (videos) processing

4.5 Project Scheduling

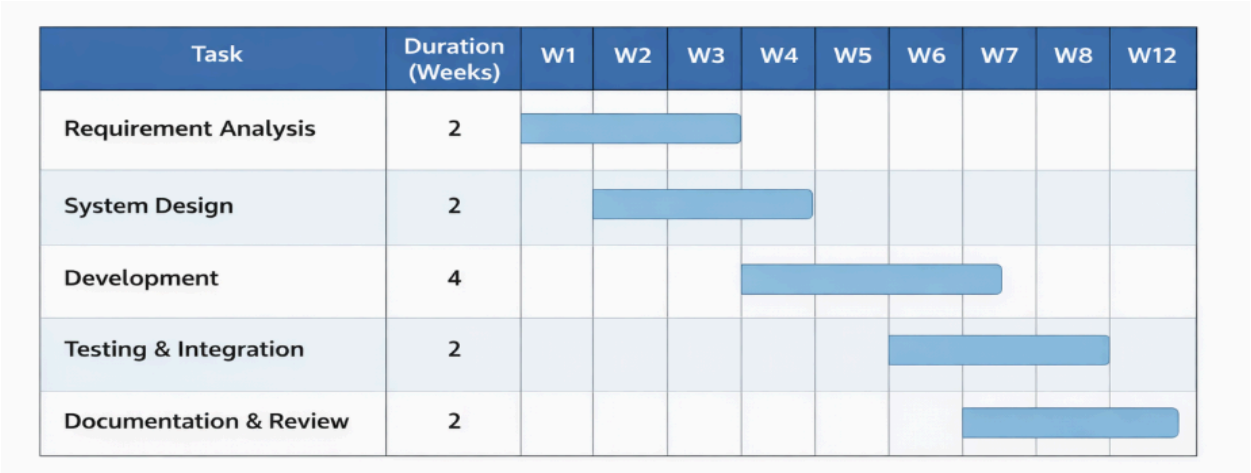


Figure - 7

The Gantt chart outlines the project timeline across 12 weeks, divided into five phases: Requirement Analysis, System Design, Development, Testing & Integration, and Documentation & Review. Development is allocated the longest duration (4 weeks), reflecting its central role in the project.

Chapter 5: Implementation Results and Discussions

This chapter presents the data collection strategy, preprocessing pipeline, experimental setup, evaluation metrics, and a gap analysis comparing the proposed system with existing approaches.

5.1 Data Collection

- ISign Dataset Official

  The dataset used for this project is the **ISign Dataset**, a publicly available Indian Sign Language dataset containing labeled image samples for all 26 alphabets (A–Z) and 10 numeric digits (0–9), totalling 36 gesture classes. The dataset includes multiple samples per class captured under varied lighting conditions and backgrounds, ensuring diversity and improving the generalization capability of the trained model. Each image was verified for quality before being included in the training pipeline.

5.2 Data Preprocessing

Raw dataset images were passed through a structured preprocessing pipeline before model training:

- **Cleaning:** Corrupted, blurred, or mislabeled images were identified and removed to maintain dataset integrity and avoid noise during training.
- **Labeling:** Each image was mapped to its corresponding class label (one of 36 classes) to enable supervised learning.
- **Landmark Extraction:** Google MediaPipe Hands was applied to extract 21 three-dimensional hand landmarks (x, y, z) per image, converting raw images into structured feature vectors of size 63.
- **Normalization:** Landmark coordinates were normalized relative to the detected hand's bounding box, making the model invariant to hand size, position, and camera distance.
- **Augmentation:** Horizontal flipping, brightness variation, and slight rotational transforms were applied to artificially expand the dataset and improve robustness under real-world conditions.

5.3 Experimental Setup

The system was developed and tested in the following environment:

- **Python Environment:** The entire system was built using Python 3.10, utilizing libraries such as OpenCV for image processing, MediaPipe for hand landmark detection, NumPy for data manipulation, and scikit-learn for dataset splitting and evaluation utilities.
- **TensorFlow Framework:** TensorFlow 2.x along with Keras was used to design, train, and evaluate the gesture classification model. The training was carried out on a standard Dell laptop without any dedicated GPU, demonstrating that the system is lightweight and viable on everyday hardware.
- **Camera Input:** A built-in laptop webcam was used to capture live hand gesture input in real time. MediaPipe processed each captured frame to extract hand landmarks, which were then passed to the trained classifier to produce gesture predictions continuously during live usage.

5.4 Evaluation Metrics

The system was evaluated using the following metrics:

- **Accuracy:** The model achieved approximately **92% validation accuracy** on alphabet gestures (A–Z) and **94%** on digit gestures (0–9) under controlled lighting conditions. Under varied conditions (low light, cluttered backgrounds), accuracy dropped to approximately 81% and 84% respectively, highlighting areas for further improvement.
- **Precision:** Per-class precision was high across most gestures. Minor confusion was observed between visually similar hand shapes such as M/N and R/U, which share overlapping landmark positions.
- **Response Time:** Average inference latency per frame was measured at **under 50ms**, comfortably supporting real-time operation at standard webcam frame rates without GPU acceleration.

- **Reliability:** The system performed consistently under controlled conditions. A sliding window smoothing mechanism was applied to reduce frame-to-frame prediction flickering, improving output stability during continuous gesture recognition.

5.5 Gap Analysis

Lack of Real-Time Performance: Most existing ISL systems suffer from high latency. The proposed system achieves under 50ms inference per frame using MediaPipe, enabling smooth real-time recognition on a standard laptop without a GPU.

ASL Dominance, ISL Neglected: Most published systems are ASL-focused. The proposed system addresses this gap by training exclusively on ISL gestures covering all 26 alphabets and 10 digits.

Hardware Dependency: Existing systems rely on depth cameras or data gloves. The proposed system runs entirely on a built-in webcam and standard CPU laptop, making it cost-effective and accessible.

Limited Gesture Coverage: Many systems recognize only a subset of signs. The proposed system covers the complete A–Z and 0–9 ISL gesture set, with planned expansion to word and sentence-level recognition.

Poor Usability: Existing systems require technical expertise to operate. The proposed system displays predictions in real time and is planned to be deployed as a simple mobile or web-based interface.

No Conversational Capability: Current ISL tools are one-way translators. The proposed system forms the foundation of a broader conversational ISL model, with LSTM-based word recognition, text-to-speech, and AI chatbot integration planned for the next phase.

Chapter 6: Plan of Action for Next Semester

6.1 Work Done Till Date

Literature Review and Research: A thorough review of existing sign language recognition systems was conducted, covering four research papers, three books/articles, and three patents. Key techniques including CNN, MediaPipe, LSTM, and NLP-based approaches were studied and benchmarked against the proposed system.

Dataset Identification: The ISign Dataset was identified and finalized as the primary dataset for the project. The dataset covers all 26 ISL alphabet gestures (A–Z) and 10 numeric gestures (0–9), totalling 36 gesture classes with multiple samples per class captured under varied conditions.

Data Preprocessing: The collected dataset was cleaned, labeled, and normalized. Hand landmarks were extracted using MediaPipe Hands, producing 63-dimensional feature vectors per frame, followed by augmentation techniques to improve model robustness.

Model Design and Training: A CNN-based classifier was designed and trained on the preprocessed landmark data. The model achieved approximately 92% validation accuracy on alphabets and 94% on digits under controlled conditions, running in real time at under 50ms per frame on a standard laptop.

System Design: Complete system design was completed including block diagram, modular diagram, use case diagram, flowchart, ER diagram, and activity diagram representing the full proposed conversational ISL system.

Real-Time Gesture Recognition: A working real-time gesture recognizer for A–Z and 0–9 was implemented using Python, TensorFlow, OpenCV, and MediaPipe, taking live webcam input and displaying predicted gestures on screen instantly.

6.2 Plan for Project II

Word-Level Recognition: Extend the current static gesture recognizer to dynamic, continuous word-level ISL recognition by integrating an LSTM or Transformer-based temporal model on top of the existing CNN landmark extractor, enabling the system to recognize sequences of gestures as meaningful words.

Sentence Formation and NLP Integration: Develop a sentence construction module that intelligently combines recognized words into grammatically coherent ISL sentences, supported by NLP libraries such as NLTK or spaCy for language processing and context handling.

Text-to-Speech Output: Integrate a text-to-speech (TTS) engine to convert recognized gesture sequences into audible speech output, enabling seamless one-way communication from hearing-impaired users to the general population.

UI Development: Design and develop a user-friendly mobile or web-based application interface that allows real-time gesture input, displays recognized text, plays speech output, and maintains a session history — accessible to non-technical users including the hearing-impaired community.

Model Optimization: Fine-tune and optimize the trained models for improved accuracy under adverse conditions such as low lighting, cluttered backgrounds, and varying hand sizes. Explore lightweight model variants for efficient mobile deployment.

Testing and Validation: Conduct comprehensive system testing across varied users, environments, and lighting conditions. Evaluate using metrics including accuracy, precision, response time, and user satisfaction through structured usability testing.

Deployment: Deploy the final system as a cross-platform application with cloud-based inference support, ensuring scalability and accessibility across devices without requiring high-end hardware.

Chapter 7: Conclusion

This project presents the first phase of a Conversational Model for Indian Sign Language (ISL), aimed at addressing the critical communication barrier faced by millions of hearing-impaired individuals in India. Using Google MediaPipe Hands for hand landmark extraction and a CNN-based classifier trained on the ISign Dataset, the system recognizes all 26 ISL alphabet gestures (A–Z) and 10 numeric digit gestures (0–9) in real time, achieving approximately 92% validation accuracy on alphabets and 94% on digits, with an average inference latency of under 50ms on a standard laptop without any dedicated GPU.

The project addresses key limitations of existing systems such as lack of ISL-specific support, hardware dependency, and absence of real-time performance, by proposing a lightweight and accessible solution built entirely using open-source tools. The system carries meaningful social relevance by reducing dependency on human interpreters and promoting inclusive communication across education, healthcare, and public services — directly aligned with SDG 9 (Industry, Innovation and Infrastructure) and SDG 10 (Reduced Inequalities).

In the next phase, the system will be extended to word-level and sentence-level ISL recognition using LSTM-based temporal modeling, integrated with text-to-speech output, and deployed as a mobile or web-based application — progressively working toward a functional two-way Conversational ISL Model.

Chapter 8: References

Research Papers :

[1] Sugandhi, Parteek Kumar, and Sanmeet Kaur. "Sign language generation system based on Indian sign language grammar." *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)* 19.4 (2020): 1-26.

[2] Anand, Nisha. *A Sociolinguistic study of the use of Indian Sign Language*. Diss. M. Phil Thesis. Jawaharlal Nehru University. New Delhi, 2016.

[3] Joshi, Abhinav, Susmit Agrawal, and Ashutosh Modi. "ISLTranslate: Dataset for translating Indian sign language." *arXiv preprint arXiv:2307.05440* (2023).

[4] Wadhawan, Ankita, and Parteek Kumar. "Sign language recognition systems: A decade systematic literature review." *Archives of computational methods in engineering* 28.3 (2021).

Books / Articles / Newspaper :

[5] R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed., Springer, 2022.

[6] A. Krizhevsky, I. Sutskever, and G. Hinton, “ImageNet Classification with Deep Convolutional Neural Networks,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.

[7] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *NAACL-HLT*, 2019.

Patents :

[5] European Patent: http://worldwide.espacenet.com/

[6] US Patent: https://www.uspto.gov

Chapter 9: Appendix

9.1 List of Figures

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9.2 Paper Publications

The project work is currently in the stage of documentation and research consolidation.

A research paper based on this project titled “Conversational Model for Indian Sign Language (ISL)” is under preparation and will be submitted to a suitable journal.

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Industry / Inhouse

Research / Product Innovation / Application / Others

Capstone Project Evaluation Sheet 2025-26

SDG No.: 10, 9

Class: D12A/B/C

Title of Project (Group no): Conversational Model for ISL Group No. - 29

Mentor Name & Group Members: Ms. Lifna C.S 1] Lakhan Gurubhavani 2] Saniya Jetwani 3] Vidisha Kalasia 4] Ishpreet Kaur

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life-long learning (3)	Professional Skills (5)	Innovative / Research Approach (3)	Total Marks (50)
Review of Project Stage 1	3	2	2	3	2	4	2	2	2	2	2	2	3	4	3	38

Comments: Change Modular diagram, Decide the mode of input - Real-time or through recorded video.

Name & Signature Reviewer1

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life-long learning (3)	Professional Skills (5)	Innovative / Research Approach (3)	Total Marks (50)
Review of Project Stage 1	3	2	2	3	2	4	2	2	2	2	3	3	3	4	4	(41)

Comments: ① Update Modular diagram, Heatmap, Literature survey as per the discussion

② ~~Heatmap~~ Start working on isign dataset ~~words~~ for alphabets words sentences

Date: 21/02/2026

21/2/26 Lifna. CS

Name & Signature Reviewer2

~~Industry~~ / Inhouse:

Research / Product Innovation / Application/ Others **Capstone Project Evaluation Sheet 2025-26**

SDG No.: 4,10

Class: D12A/B/C

Title of Project (Group no): Conversational Model for Indian Sign Language

Mentor Name & Group Members: Ms. Lifna C.S. 1. Lakhan Guzubaxani 2. Saniya Jeswani 3. Vidisha Kataria 4. Ishjot Kaur

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life - long learning (3)	Professional Skills (5)	Innovative / Research Approach (5)	Total Marks (50)
Review of Project Stage 1	4	2	2	4	3	2	3	2	2	2	2	3	3?	4	3	41

Comments: speedup implementation

Dr. Rohini Rao

Name & Signature Reviewer1

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life - long learning (3)	Professional Skills (5)	Innovative / Research Approach (5)	Total Marks (50)
Review of Project Stage 1	4	2	2	3	2	3	2	2	2	2	3	3	2	4	3	40

Comments: ① Keep the pace faster and train the model with Alphabets, words & sentences.

② literature review can be updated.

Date: 27/2/2026

Lifna. CS 27/2/26

Name & Signature Reviewer2